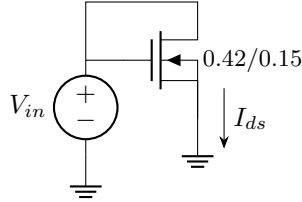


EC 5311 Digital IC Design: Assignment 2
Transfer characteristics of transistors

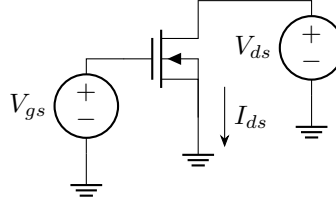
$$\begin{array}{llll}
 V_{Tn} = 0.7 \text{ V}, & \mu_n = 0.025 \text{ m}^2/\text{Vs}, & C_{oxn} = 8.34 \text{ fF}/\mu\text{m}^2, & v_{satn} = 8 \times 10^4 \text{ m/s}, & \lambda_n = 0.2/\text{V} \\
 |V_{Tp}| = 0.7 \text{ V}, & \mu_p = 0.009 \text{ m}^2/\text{Vs}, & C_{oxp} = 8.16 \text{ fF}/\mu\text{m}^2, & v_{satp} = 3 \times 10^4 \text{ m/s}, & \lambda_p = 0.2/\text{V}
 \end{array}$$

1. (a) Consider the nMOS transistor connected as shown below.



Is it in the saturation or linear region? Using ngspice, obtain the transfer characteristics I_{ds} vs V_{gs} assuming $L = 0.15\mu\text{m}$ and $W = 0.42\mu\text{m}$. Plot the simulated model and analytical model in the same plot and find the mean percentage error.

- (b) Connect sources to the nMOS transistor as shown and obtain the output characteristics for $V_{gs} = 0.6\text{V}, 1\text{V}, 1.4\text{V}$ and 1.8V . Plot the analytical model and obtain the mean percentage error.



- (c) Simulate the I_{ds} vs V_{ds} for $V_{gs} = V_{DD}$ for transistors of different W/L keeping W/L ratio constant while increasing the sizes. Note how the velocity saturation effect decreases as overall length increases.
2. Repeat the exercise for the pMOS transistor.